

Lab 5 – Strings and Static Methods 2

Aim

This lab aims to reinforce your use of methods and strings to make you think about problem solving.

Tips:

1. Some may be challenging problems. Please work with your **Lab Group mates**.
2. Try to think about the design and the logic of this program before you code it. It may help to think about it on paper, and use Java visualizer <https://pythontutor.com/java.html>
3. Build this program slowly. Get each step working one at a time, print the intermediate results to test incrementally.

Lab Task 5.1 Repeat Strings

- We want to write a function/static method `repeatString` that given an input string, returns a new string made of repetitions of that input string.
- The static method takes the input string as well as how many times the input string needs to be repeated.
- Test cases:
 - `repeatString("abc", 3)` → `"abccabccabc"`
 - `repeatString("abcd", 1)` → `"abcd"`
 - `repeatString("abcd", 0)` → `""`

```
run:
abccabccabc
abcd

BUILD SUCCESSFUL (total time: 0 seconds)
```

- Create a new project Lab6.
- Create a new class `RepeatString`.
- Create a main method.
- Create an empty static method `repeatString`:
 - Write the method header
`public static String repeatString(String str, int count) {`
 - To make the compile error go away, make it so that it returns an empty string for now
- Complete the main method with the given test cases above, for example:
 - `System.out.println(repeatString("abc", 3));`
- Add more test cases of your own
- Now, complete the body of `repeatString` method, and test it against the testcases

- Hint:
 - Start with a String local variable `result` and initialize it with an empty string
 - Use a for-loop to repeat `count` times: concatenate using operator `+` the variable `result` with the input string `str`, and store it back to `result`

Lab Task 5.2 Counting Question Marks

- We want to write a static method `countQuestionMark` that given an input string, returns how many times the substring `"?"` (or equivalently, the character `'?'`) appears in the input string.
- Test cases:
 - `countQuestionMark("ab???ab")` → 3
 - `countQuestionMark("?abc?")` → 2
 - `countQuestionMark("")` → 0

```
run:
3
2
0
BUILD SUCCESSFUL (total time: 0 seconds)
```

- Create a new class `CountQuestionMark`.
- Create a main method.
- Create an empty static method `countQuestionMark`:
 - Write the method header
`public static int countQuestionMark(String str) {`
 - To make the compile error go away, make it so that it returns 0 for now
- Complete the main method with the given test cases above, for example:
 - `System.out.println(countQuestionMark("ab???ab"));`
- Add more test cases of your own
- Now, complete the body of `countQuestionMark` method, and test it against the testcases
- Hint:
 - Store the counting result with an `int` local variable `count` and initialize it to 0
 - Use a for-loop to iterate with `int i`, `str.length()` times:
 - Testing string equality by using an `if` statement, `substring` and `equals` methods
 - Recall that `equals` method is used to compare objects, in this case, `String` objects
 - To extract the substring to be tested, the `substring` method should start with index `i` and end with but not including index `i+1`
 - Return the counting result
- Next, treat the question mark as a character `'?'`

- Comment out the testing string equality part
- Use the method `charAt(i)` to extract a character from the string at index `i`
- Use operator `==` to compare characters

Lab Task 5.3 Check Membership

- We complete the problem introduced at the end of Lecture 5, rewritten below.
- Write a method to validate a membership card id, with the following conditions:
 - Must be 13 characters long
 - Must be 4 blocks
 - Must be separated by hyphens
 - First character must be between A and D
 - can be upper or lower case
 - Other 3 blocks are 3 characters long
 - 2nd and 3rd blocks are all numbers
 - Final block must be 2 numbers and a letter
 - the letter must be between A and T
- Test cases:
 - `checkMembership("A-123-456-23A")` → true
 - `checkMembership("A-187-267-111")` → false
 - `checkMembership("c-542-223-11G")` → true
 - `checkMembership("A-187-267+111")` → false
 - `checkMembership("A-187-11G")` → false
 - `checkMembership("E-123-456-23A")` → false
 - `checkMembership("b-555-88-123T")` → false

```
63 public static void main(String[] args) {
64     String test1 = "A-123-456-23A";
65     boolean valid = checkMembership(test1);
66     System.out.println("Case 1 Expect: true Real: " + valid);
67
68     String test2 = "A-187-267-111";
69     valid = checkMembership(test2);
70     System.out.println("Case 2 Expect: false Real: " + valid);
71
72     String test3 = "c-542-223-11G";
73     valid = checkMembership(test3);
74     System.out.println("Case 3 Expect: true Real: " + valid);
75 }
```

Output - Lab6 (run) x

```
run:
Case 1 Expect: true Real: true
Case 2 Expect: false Real: false
Case 3 Expect: true Real: true
Case 4 Expect: false Real: false
Case 5 Expect: false Real: false
Case 6 Expect: false Real: false
Case 7 Expect: false Real: false
Case 8 Expect: false Real: false
Case 9 Expect: false Real: false
BUILD SUCCESSFUL (total time: 0 seconds)
```

This is the end of CPT111 2324 Lab 5 Task Sheet.